

CS & IT ENGINEERING

COMPILER DESIGN

SYNTAX DIRECTED TRANSLATION

Lecture No. -

1



By- Venkat sir

Recap of Previous Lecture



Topic

Semantic Analysis

Topic

Syntax Directed translation Semantic errors

Topics to be Covered



Topic

Syntax Directed translation

Topic

??

Topic

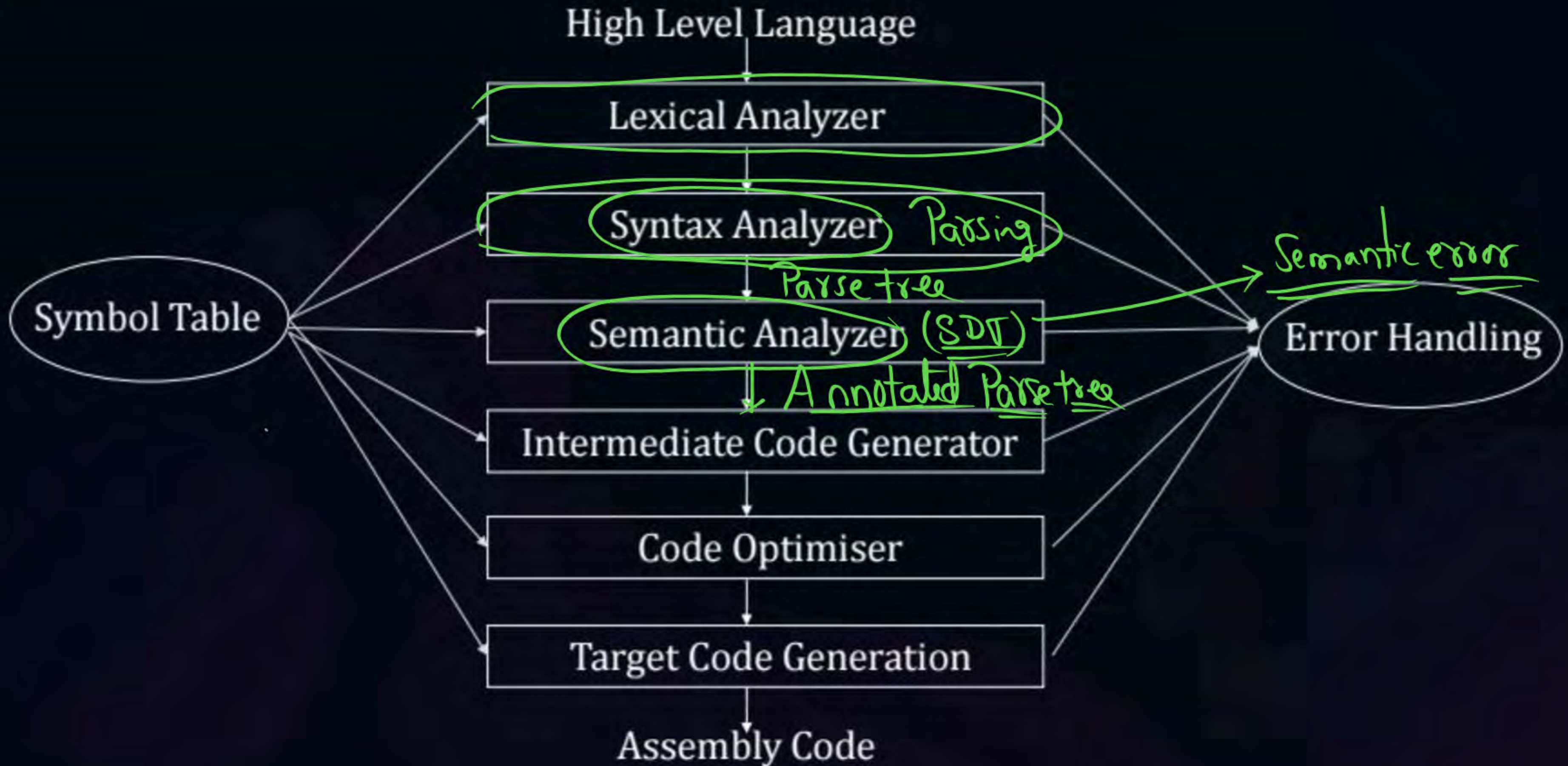
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B.V.P { ① LR Parsing
② Operator precedence Parsing

T.D.P { ③ LL(1) Parsing
④ Recursive Descent Parsing

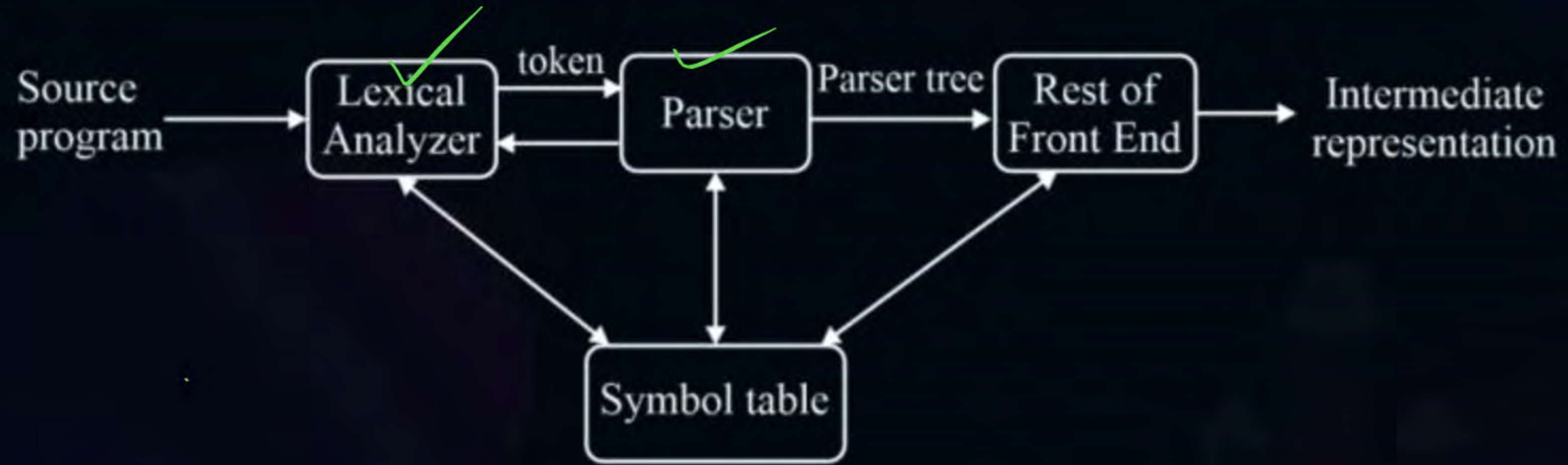


Topic : Introduction of Compiler Design





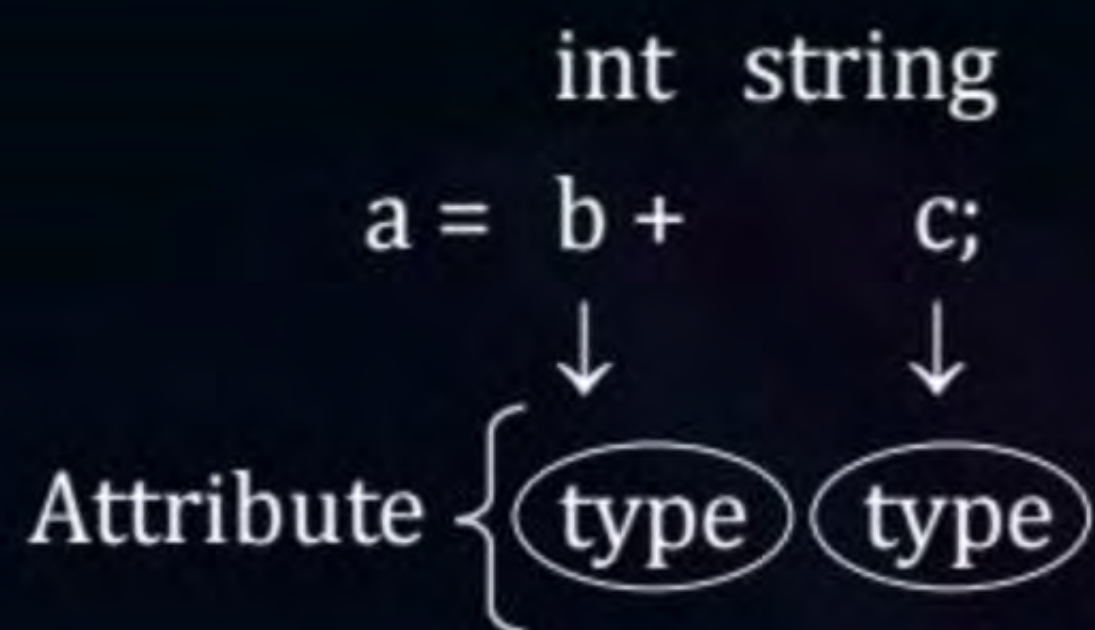
Topic : Syntax Directed Translation





Topic : Syntax Directed Translation

Semantic:- Meaning Verification





Topic : Syntax Directed Translation

Attribute Grammar:-

Attribute Grammar is a CFG where each grammar symbol if associated symbol is associated with attribute information Attribute can be type of a variable (or) Scope of a Variable.



Topic : Syntax Directed Translation

SDT = CFG + Semantic rules

$$S \rightarrow id_{type} = E_{type} \quad \{E.type = id.type\}$$
$$E \rightarrow E_{type} = E_{type} \quad \{E.type = E_1.type + E_2.type\}$$
$$E \rightarrow id_{type} \quad \{E.type = id.type\}$$



Topic : Syntax Directed Translation

Syntax Directed Translation:-

Every CFG production is associated with set of semantic rules.

Semantic rules are attribute computation rules in annotated Parse tree.

Attribute, type

$a+b$
int string

$\left\{ \begin{array}{l} S \rightarrow AB \\ A \rightarrow a \\ B \rightarrow b \end{array} \right\}$



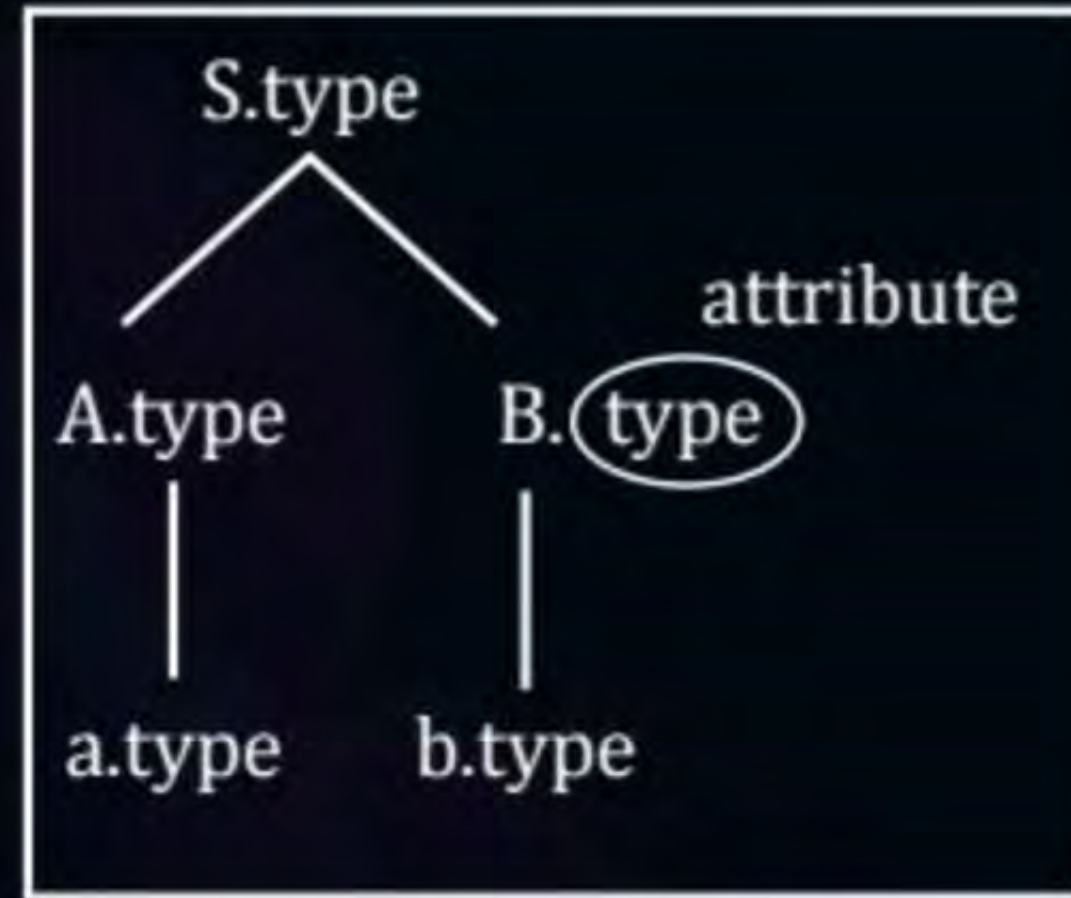


Topic : Syntax Directed Translation

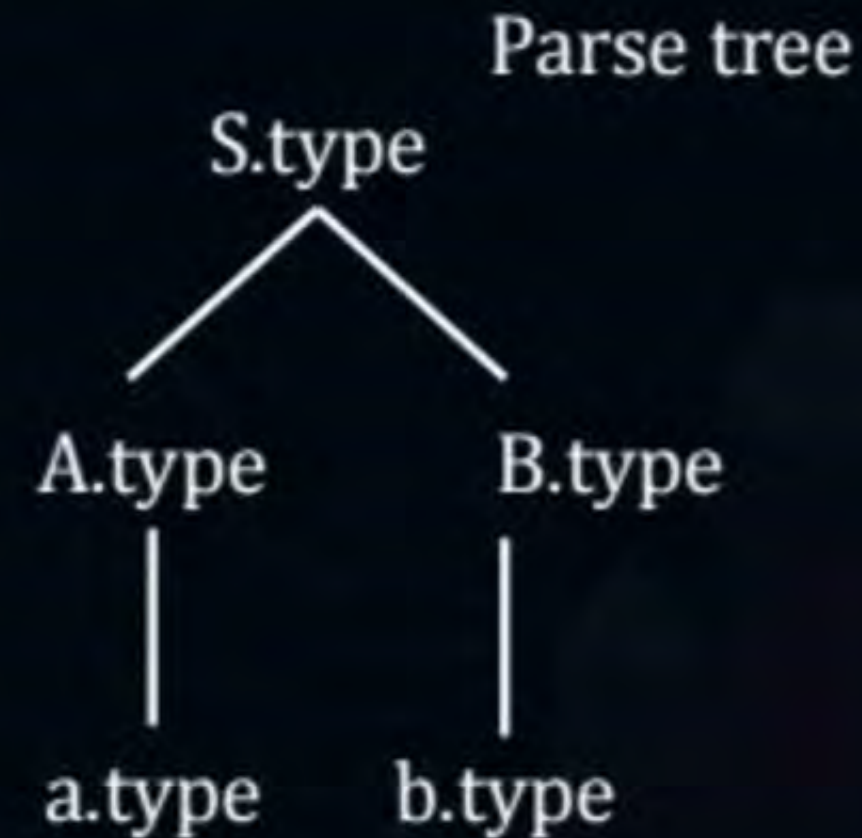
Annotated Parse tree:-

It is a parse tree in which each grammar symbol is associated with attribute information.

{ Annotated
Parse
Tree }



Decorated parse tree



Parse tree



Topic : Syntax Directed Translation

SDT Advantages:-

- (1) SDT is used to construct semantic Analysis.
- (2) SDT is used to construct Intermediate code generation.
- (3) SDT is used to construct symbol Table.
- (4) SDT is used to construct syntax tree.



Topic : Syntax Directed Translation

Steps to construct SDT:-

- (1) Construct Parse tree.
- (2) Construct annotated parse tree
- (3) Compute attribute information
- (4) Generalize rules and attach to grammar production.



Topic : Syntax Directed Translation

#Q. Consider the following SDT

1. $S \rightarrow aA \rightarrow \{\text{print}(1)\}$
2. $S \rightarrow a \rightarrow \{\text{print}(2)\}$
3. $A \rightarrow Sb \rightarrow \{\text{print}(3)\}$

This SDT if carried out with BUP what is output for input aab.

Reduce $\Rightarrow$ Semantic

231

B.U.P

Output

$\begin{array}{c} S \quad a \quad b \\ \uparrow \\ x \quad a \quad a \quad b \end{array}$

$\begin{array}{c} S \\ a \quad A \\ a \quad Sb \quad 2 \\ \uparrow \\ a \quad a \quad b \quad 1 \end{array}$



Topic : Syntax Directed Translation

$SDT = \{CFG + \text{Semantic Rule}\}$

$S \rightarrow AS \{ \text{print (1)} \}$

$S \rightarrow AB \{ \text{print (2)} \}$

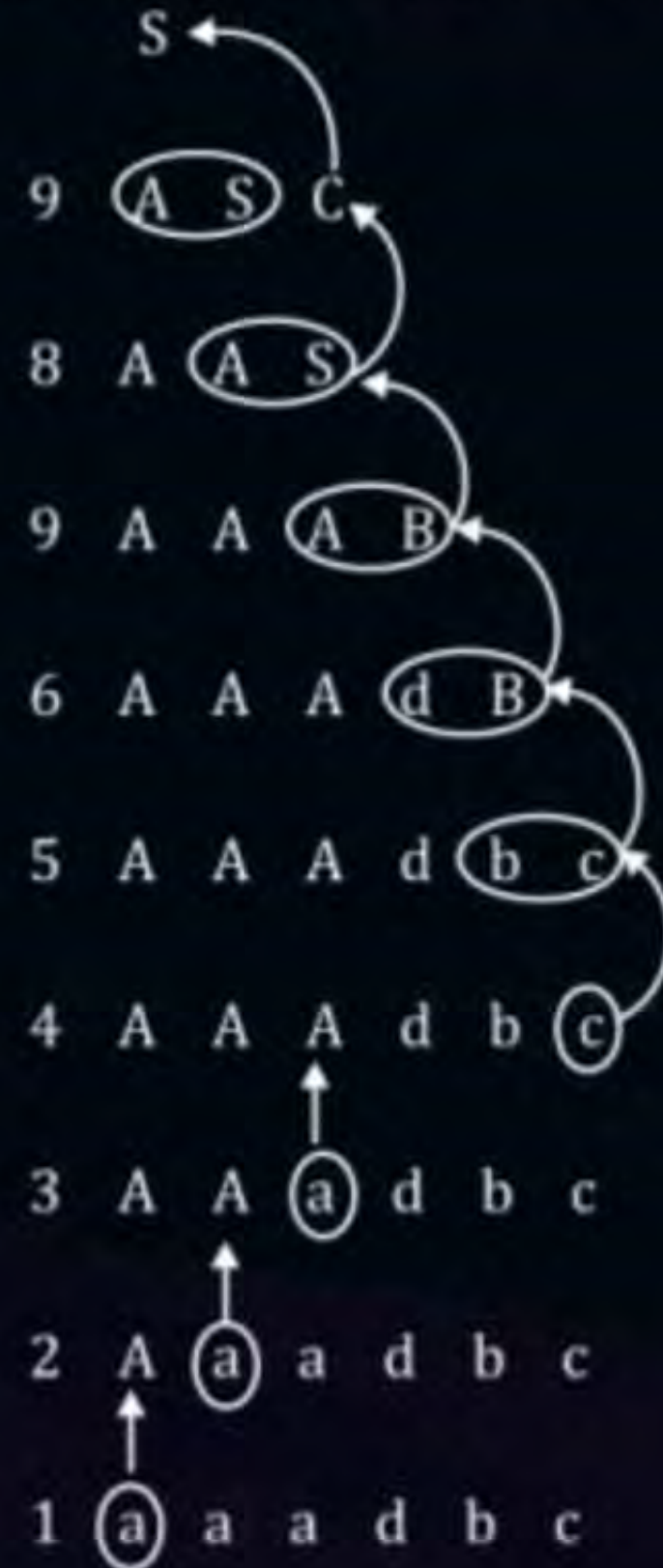
$A \rightarrow a \leftrightarrow \{ \text{print (3)} \}$

$B \rightarrow bC \{ \text{print (4)} \}$

$B \rightarrow bB \{ \text{print (5)} \}$

$C \rightarrow c \{ \text{print (6)} \}$

This SDT is carried out with Bottom up Parse what is the



Aaadbcb.

333645211

Output



Topic : Syntax Directed Translation

$\{SDT = \{CFG\} + \{\text{Semantic Rule}\}\}$

$E \rightarrow E \# T \quad \{E.val = E_1.val \# T.val\}$

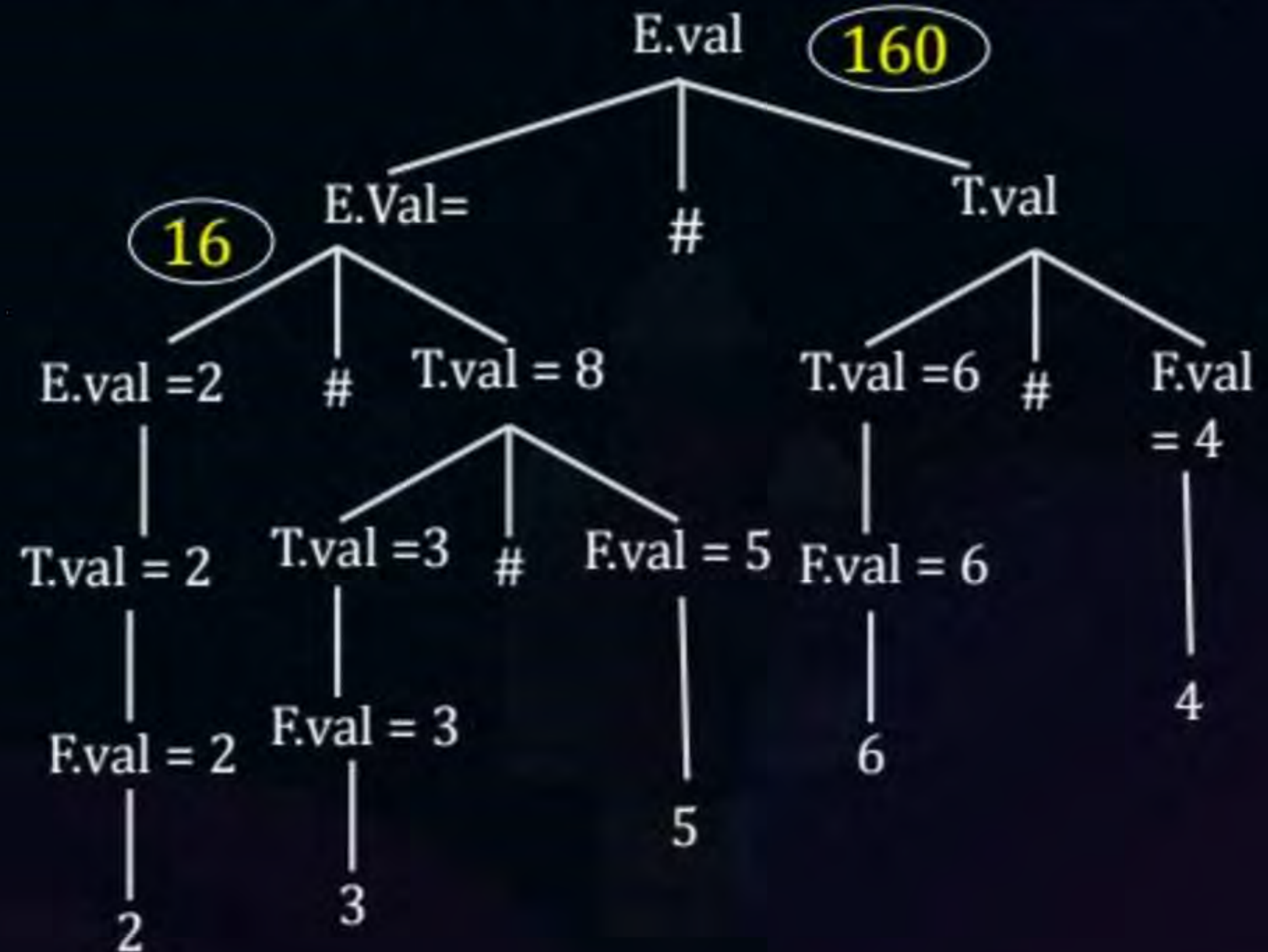
$E \rightarrow T \quad \{E.val = T.val\}$

$T \rightarrow T \& F \quad \{T.val = T_1.val + F.val\}$

$T \rightarrow F \quad \{T.val = F.val\}$

$F \rightarrow \text{num} \quad \{F.val = \text{num.val}\}$

Compute E.val for the root for the following expression $2 \# 3 \& 5 \# 6 \& 4$





Topic : Syntax Directed Translation

$\{SDT = \{CFG\} + \{\text{Semantic Rule}\}\}$

$E \rightarrow E_1 \# T \{E.val = E_1.val \# T.val\}$

$E \rightarrow T \{E.val = T.val\}$

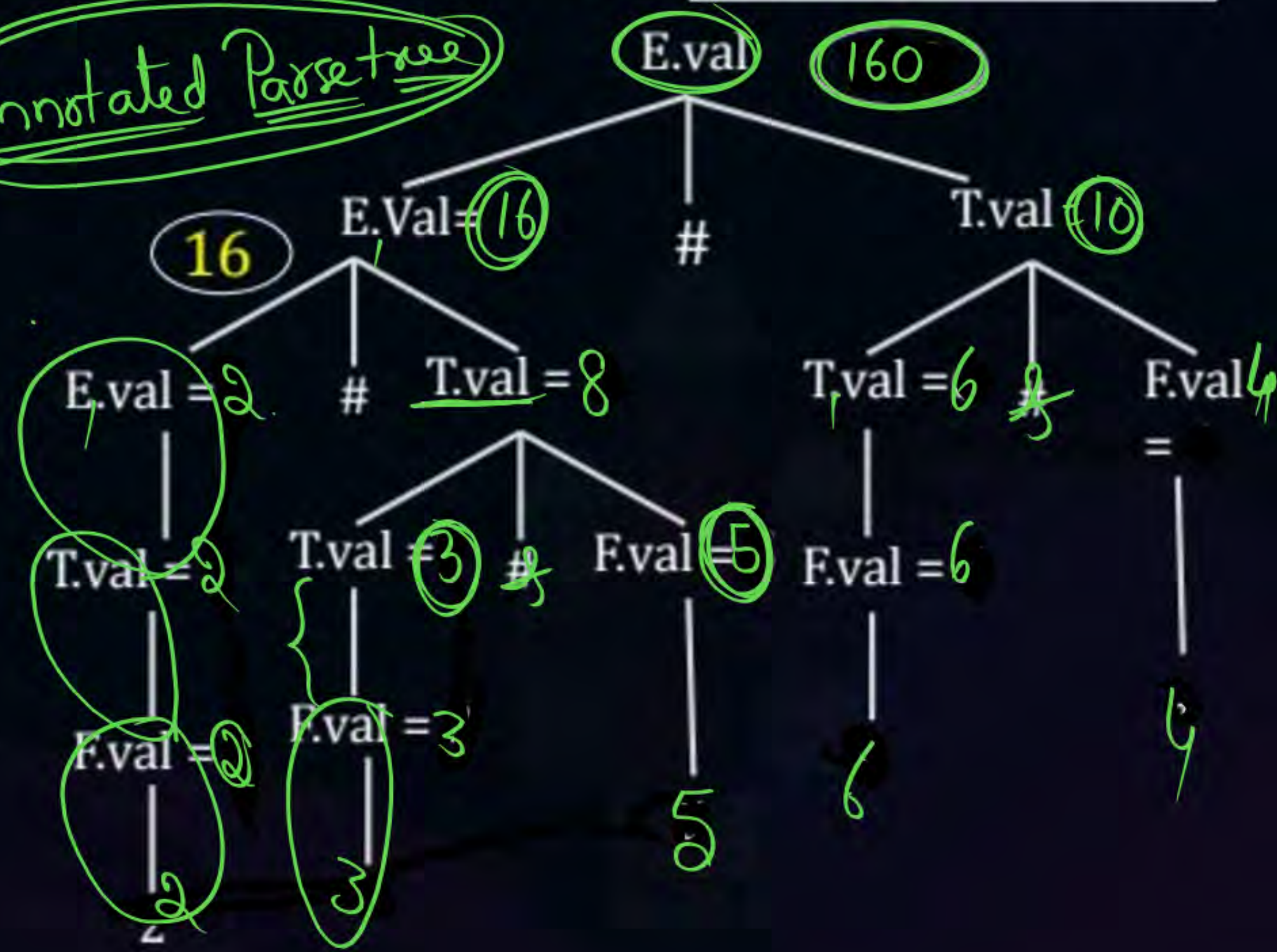
$T \rightarrow T \& F \{T.val = T_1.val + F.val\}$

$T \rightarrow F \{T.val = F.val\}$

$F \rightarrow \text{num} \{F.val = \text{num.val}\}$

Compute $E.val$ for the root for the following expression $2 \# 3 \& 5 \# 6 \& 4$

Annotated Parse tree



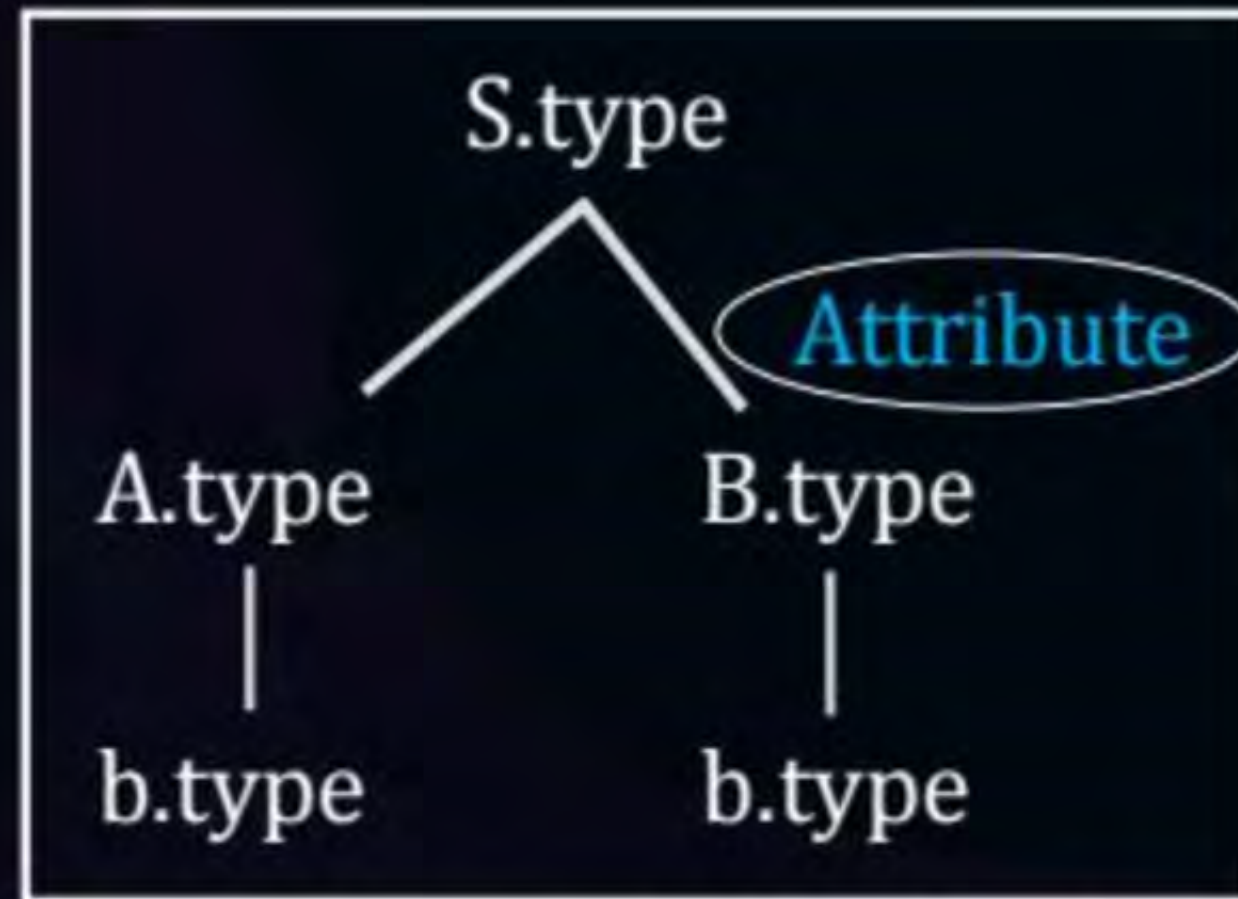


Topic : Syntax Directed Translation

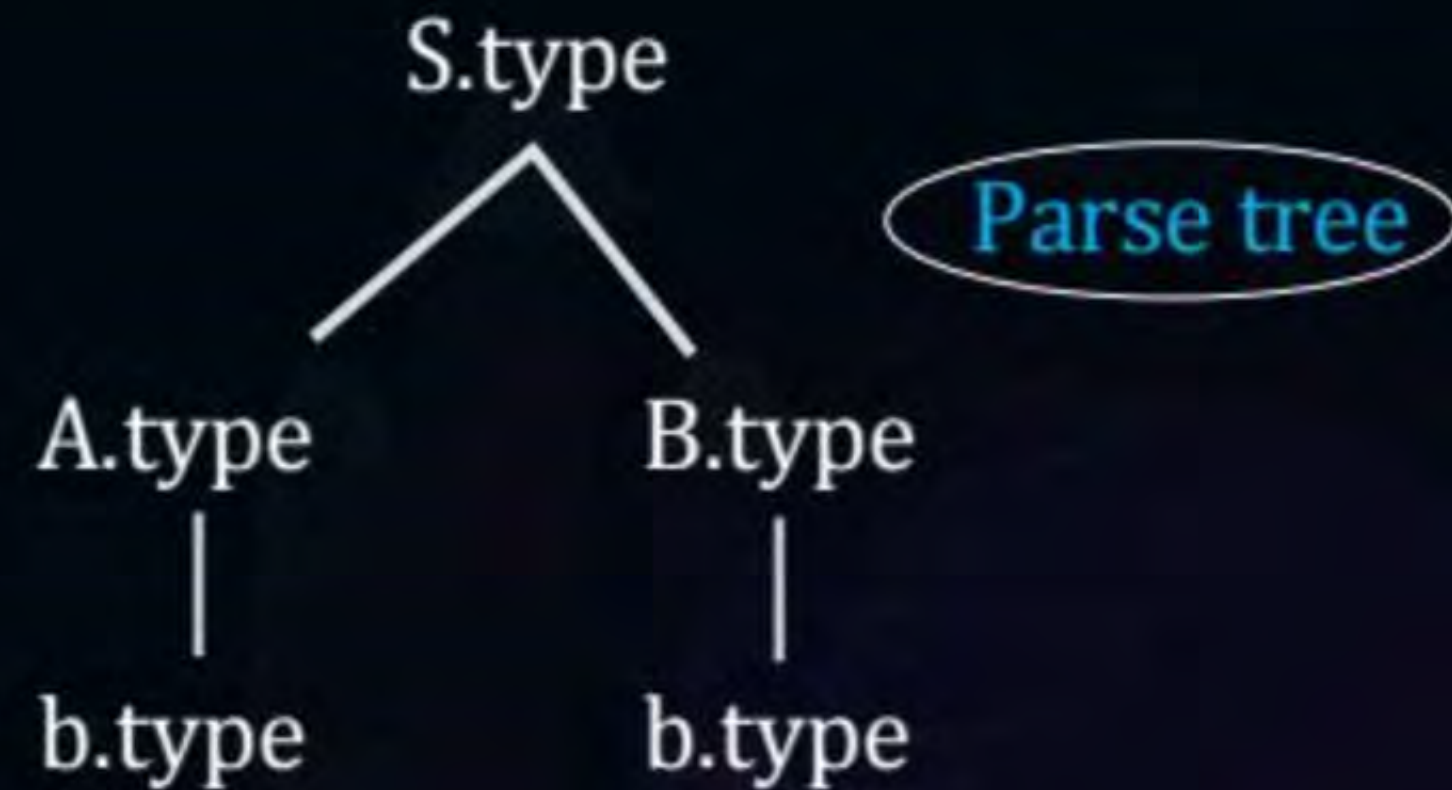
Annotated Parse tree:-

It is a parse tree in which each grammar symbol is associated with attribute information.

{ Annotated
Parse tree }



Decorated parented





Topic : Syntax Directed Translation

SDT Advantages:-

1. SDT is used to construct semantic Analysis
2. SDT is used to construct Intermediate code generation
3. SDT is used to construct symbol table.
4. SDT is used to contract syntax tree.



Topic : Syntax Directed Translation

Steps to construct SDT:-

1. Construct Parse tree
2. Construct annotated parse tree
3. Compute attribute information
4. Generalize rule and attach to grammar production.



Topic : Syntax Directed Translation

$S \rightarrow AS \{ \text{print (1)} \}$

$S \rightarrow AB \{ \text{print (2)} \}$

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$B \rightarrow bB \{ \text{print (5)} \}$

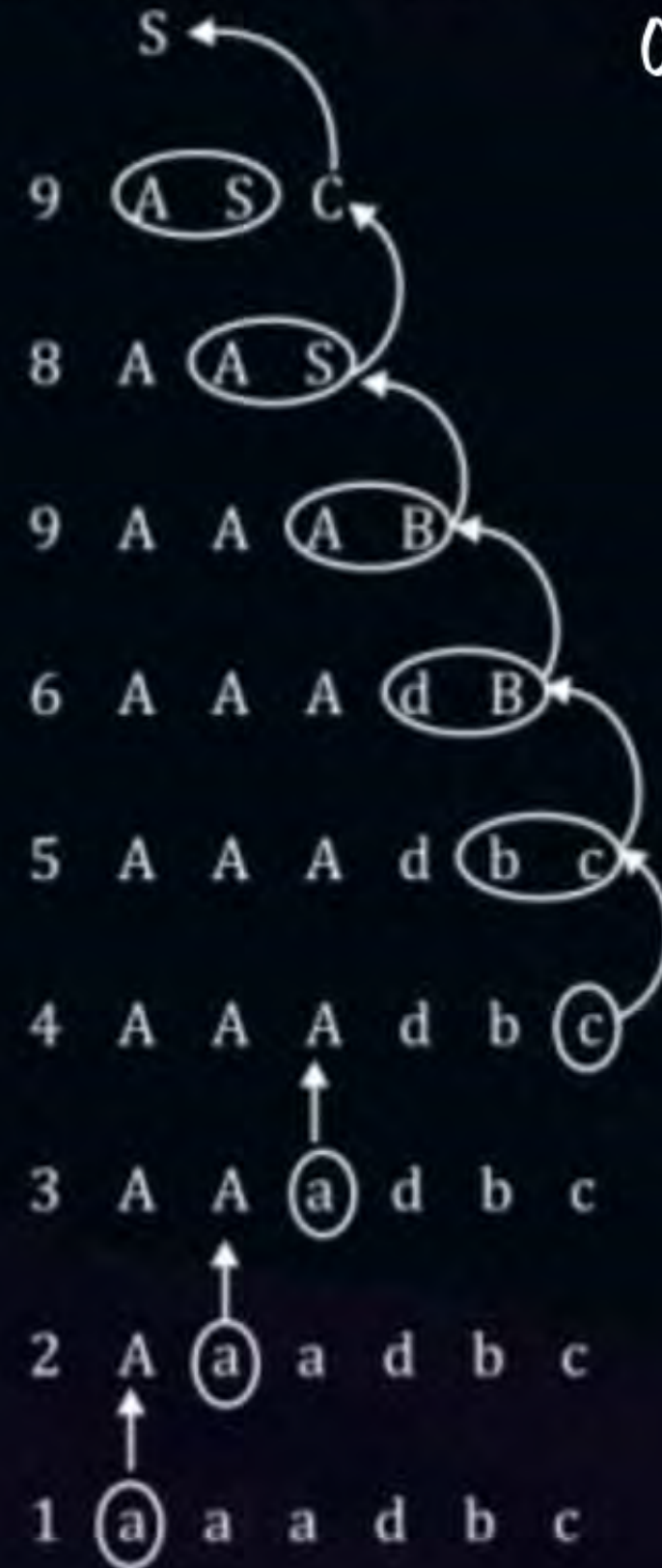
$C \rightarrow c \{ \text{print (6)} \}$

This SDT is carried out with Bottom up Parse what is the

output for input
aaadbc

333645211

Output



THANK - YOU